

Subject: Mathematics – II(303191151)

Semester: 2nd Sem. B.Tech Programme (All Branches)

Lecture Note : Unit – 2 Power Series

Series Solutions of Differential Equations

Introduction

In mathematics, the **Power Series Method** is used to seek a power series solution to certain differential equations. In general, such a solution assumes a power series with unknown coefficients, then on substituting that solution into the differential equation to find a recurrence relation for the coefficients. In this way a power series solution is obtained.

In this unit, we will learn two methods to obtain power series solution.

(i) *Power series method*

(ii) *Frobenious Method.*

Standard form of the Differential Equation

Consider a homogeneous linear second order differential equation with variable coefficients

$$P_0(x) \frac{d^2y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x)y = 0$$

Here we develop the method of solving equations of the type

$$P_0(x) \frac{d^2y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x)y = 0 \dots \dots \dots (1)$$

Where $P_0(x), P_1(x)$ and $P_2(x)$ are polynomials in x in terms of an infinite convergent series.

Assuming $P_0(x) \neq 0$, dividing (1) by $P_0(x)$, we have

$$\frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0 \dots \dots \dots (2)$$

$$\text{where } P(x) = \frac{P_1(x)}{P_0(x)} \text{ and } Q(x) = \frac{P_2(x)}{P_0(x)}$$

The power series method is the standard basic method for solving linear differential equations with variable coefficients. It gives solution in the form of power series.

Definition:-Power Series

A power series in power of $(x - x_0)$ is an infinite series of the form

$$\sum_{k=0}^{\infty} a_k (x - x_0)^k = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \dots (3)$$

where $a_0, a_1, a_2, \dots$ are constants, called the coefficients of the series x_0 is a constant called the centre of the series and x is a variable. If in particular $x_0 = 0$ we obtain a power series in power of x .

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots \dots \dots (4)$$

Existence of Power Series solutions

Every differential equation of the form

$$P_0(x) \frac{d^2 y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x)y = 0 \dots \dots \dots (1)$$

does not have series solution Assuming $P_0(x) \neq 0$, the above equation is written in the standard form as

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0 \dots \dots \dots (2)$$

$$\text{where } P(x) = \frac{P_1(x)}{P_0(x)} \text{ and } Q(x) = \frac{P_2(x)}{P_0(x)}$$

The behaviour of solutions of (2) near a point x_0 depends on the behavior of its coefficient functions $P(x)$ and $Q(x)$ near this point x_0 .

Classification of Singularities

Definition: Analytic

A function $f(x)$ is said to be analytic at x_0 if $f(x)$ has Taylor's series expansion about x_0 given by

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(x_0)}{n!} (x - x_0)^n$$

Exists and converges to $f(x)$ for all x in some open interval including x_0 . If a function $f(x)$ is not analytic at x_0 then it is called Singular at x_0 .

Definition: Ordinary Point

A point $x = x_0$ is said to be an ordinary point of differential equation (2) if both $P(x)$ and $Q(x)$ are analytic at x_0 ; that is, if both $P(x)$ and $Q(x)$ have Taylor Series representations but $x = x_0$

Definition: Singular Point

A point x_0 is said to be a singular point of (2) if either $P(x)$ or $Q(x)$ or both are not analytic at x_0

(OR) A point $x = x_0$ that is not an ordinary point of (1) is called a singular point.

Definition: Regular Singular Point (RSP)

A point $x = x_0$ of the equation

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0 \dots \dots \dots (2)$$

Is said to be regular singular point if both the following terms are analytic at x_0 .

$$(i) (x - x_0)P(x) \quad (ii) (x - x_0)^2 Q(x)$$

NOTE: If either of the above terms or both are not analytic at x_0 , then x_0 is called **An Irregular Singular Point**.

Question 1. Find ordinary point, singular point of given below two equations.

$$(i) (1 - x^2) \frac{d^2y}{dx^2} - 6x \frac{dy}{dx} - 4y = 0$$

Solution: $P(x) = \frac{P_1(x)}{P_0(x)} = \frac{-6x}{(1-x^2)}$ and $Q(x) = \frac{-4}{(1-x^2)}$.

In this example the points $x = -1$ and $x = 1$ are singular points of the equation. Except $x = -1$ and 1 all other points are ordinary points.

$$(ii) (x^2 + 4) \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 12y = 0$$

Solution: $P(x) = \frac{P_1(x)}{P_0(x)} = \frac{2x}{(4+x^2)}$ and $Q(x) = \frac{-12}{(4+x^2)}$

It has singular points $x = \pm 2i$.

Other than $x = \pm 2i$ all other points are ordinary points.

Question 2. Find singular points and classify them into regular singular point or irregular singular point.

$$(i) x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$$

Solution: The point $x = 0$ is a singular point of the equation. Since the factor $(x - 0)$ occurs to only the first power in the denominator of $P(x) = -\frac{x}{x^2} = -\frac{1}{x}$ and to only the second power in $Q(x) = -\frac{3}{x^2}$, we conclude that $x = 0$ is a regular singular point

$$(ii) 2x(x-2)^2 y'' + 3xy' + (x-2)y = 0$$

Solution: $P(x) = \frac{3}{2(x-2)^2}$, $Q(x) = \frac{1}{2x(x-2)}$

$P(x)$ or $Q(x)$ or both are infinite at $x = 0$ and 2 . They are not analytic at $x = 0$ and 2 .

Thus $x = 0$ and 2 are singular point of the equation.

Now, $xP(x) = \frac{3x}{2(x-2)^2}$ and $x^2Q(x) = \frac{x}{2(x-2)}$

Thus both $xP(x)$ and $x^2Q(x)$ are analytic at $x = 0$.

The point $x = 0$ is a regular singular point.

Now, $(x-2)P(x) = \frac{3}{2(x-2)}$ and $(x-2)^2Q(x) = \frac{x-2}{2x}$

Thus $(x-2)P(x)$ is an irregular singular point.

$$(iii) x(x+1)^2 y'' + (2x-1)y' + x^2 y = 0$$

Solution: $P(x) = \frac{2x-1}{x(x+1)^2}$, $Q(x) = \frac{x}{(x+1)^2}$

Since $P(x)$ or $Q(x)$ or both are undefined at $x = 0$ and $x = -1$, they are singular point of the given equation.

Again $(x-0)P(x) = \frac{2x-1}{x(x+1)^2}$ and $(x-0)^2Q(x) = \frac{x^3}{(x+1)^2}$

$(x-0)P(x)$ and $(x-0)^2Q(x)$ are analytic at $x=0$. Thus $x=0$ is a regular singular point.

Again $(x+1)P(x) = \frac{2x-1}{x(x+1)}$, $(x+1)^2Q(x) = x^3$

But $(x+1)P(x)$ is not analytic at $x = -1$. Thus $x = -1$ is an irregular singular point.

POWER SERIES SOLUTION NEAR AN ORDINARY POINT

Let $x = 0$ be an ordinary point of the equation

$$P_0(x) \frac{d^2 y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x)y = 0 \dots \dots \dots (i)$$

$$\text{Or } \frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

$$\text{where } P(x) = \frac{P_1(x)}{P_0(x)} \text{ and } Q(x) = \frac{P_2(x)}{P_0(x)}, (P_0(x) \neq 0).$$

Let a solution of (i) be given as

$$y = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots \dots \dots (ii)$$

$$\therefore \frac{dy}{dx} = \sum_{k=1}^{\infty} k a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$\text{and } \frac{d^2y}{dx^2} = \sum_{k=2}^{\infty} k(k-1)a_k x^{k-2} = 2a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots$$

Substituting these values in (i), and equating coefficients of various powers of x to 0.

Equating the coefficient of x^n , we obtain the recurrence relation.

Assigning different values to n in this recurrence relation, we can determine the unknown coefficients, in (ii) successively and a_i 's in terms of a_0 and a_1 .

Using these values of a_i 's in (i) we can obtain series solution of given differential equation.

EXAMPLE:-1: Solve the equation $y' - y = 0$ by the power series method.

Solution: given equation

$$y' - y = 0 \dots \dots \dots (i)$$

Differentiating the power series

$$y = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots \dots \dots (ii)$$

term by term, we get

$$\therefore y'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots \dots \dots (iii)$$

writing (i) as $y' = y$ and comparing (iii) with, we get

$$a_1 = a_0, \quad 2a_2 = a_1, \quad 3a_3 = a_2 \dots \dots \dots \text{and so on.}$$

In general we have

$$(k+1)a_{k+1} = a_k, \quad k = 0, 1, 2, 3 \dots$$

Therefore we can express a_{k+1} in terms of a_k as

$$a_{k+1} = \frac{1}{k+1} a_k, \quad k = 0, 1, 2, 3 \dots$$

Let us compute the first few coefficients explicitly

$$a_1 = \frac{1}{1} a_0, \quad a_2 = \frac{1}{2} a_1 = \frac{1}{2 \cdot 1} a_0, \quad a_3 = \frac{1}{3} a_2 = \frac{1}{3 \cdot 2 \cdot 1} a_0 \dots$$

From here it is clear that

$$a_k = \frac{1}{k!} a_0$$

$$y(x) = \sum_{k=0}^{\infty} \frac{1}{k!} a_0 x^k = a_0 \sum_{k=0}^{\infty} \frac{x^k}{k!} = a_0 \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \right) = a_0 e^x$$

EXAMPLE:-2: Solve the equation $\frac{d^2 y}{dx^2} + y = 0$ by the power series method.

Solution:

Let the series solution be

$$y(x) = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots \dots \dots (i)$$

$$\therefore \frac{dy(x)}{dx} = \sum_{k=1}^{\infty} k a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$\text{and } \frac{d^2 y(x)}{dx^2} = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} = 2a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots$$

Substituting in given equation, we get

$$\sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} + \sum_{k=0}^{\infty} a_k x^k = 0.$$

$$\text{i.e. } (2a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots) + (a_0 + a_1 x + a_2 x^2 + \dots) = 0$$

$$\text{i.e. } (2a_2 + a_0) + (3 \cdot 2 a_3 + a_1)x + (4 \cdot 3 a_4 + a_2)x^2 + \dots$$

$$+ ((n+2)(n+1)a_{n+2} + a_n)x^n + \dots = 0$$

Comparing the coefficients, we get

$$2a_2 + a_0 = 0,$$

$$3 \cdot 2 a_3 + a_1 = 0,$$

$$4 \cdot 3 a_4 + a_2 = 0,$$

...

$$(n+2)(n+1)a_{n+2} + a_n = 0$$

Solving these equations, we get

$$a_2 = -\frac{a_0}{2!}, a_3 = -\frac{a_1}{3!}, a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}, \dots$$

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)} \dots \dots \dots (ii)$$

Using (ii), for $n = 3$ $a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}$

Substituting these values in (i), we get

$$y = a_0 \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) + a_1 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right)$$

Which is the required series solution of given differential equation.

Note: For above solution, considering the Maclaurin's series:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots; \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

We get

$$y = a_0 \cos x + a_1 \sin x$$

EXAMPLE:2. Solve the differential equation $(1 - x^2)y'' - 2xy' + 2y = 0$ using power series method.

Solution:

Here $x = 0$ is an ordinary point, and except $x = -1$ and $x = 1$ all other points are regular points.

Let the series solution be

$$y = \sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots \dots \dots (i)$$

$$\therefore y' = \sum_{k=1}^{\infty} k a_k x^{k-1} = a_1 + 2a_2 x + 3a_3 x^2 + \dots$$

$$\text{and } y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} = 2a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots$$

Substituting in given equation, we get

$$(1 - x^2) \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} - 2x \sum_{k=1}^{\infty} k a_k x^{k-1} + 2 \sum_{k=0}^{\infty} a_k x^k = 0.$$

$$\text{i.e. } \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2} - \sum_{k=0}^{\infty} [k(k-1) + 2k - 2] a_k x^k = 0$$

Collecting and comparing the coefficient of x^n , we get

$$(n+2)(n+1)a_{n+2} = (n+2)(n-1)a_n$$

$$(n+1)a_{n+2} = (n-1)a_n$$

$$a_{n+2} = \frac{n-1}{n+1} a_n$$

For $n = 0, a_2 = -a_0$,

$n = 1, a_3 = 0$, which gives $a_3 = a_5 = a_7 = a_9 = \dots = 0$

Now, for even values of n , let $n = 2m$.

$$\therefore a_{2m+2} = \frac{2m-1}{2m+1} a_{2m} \text{ for } m = 0, 1, 2, \dots$$

For $m = 0, a_2 = -a_0$

For $m = 1, a_4 = \frac{1}{3} a_2 = -\frac{1}{3} a_0$,

For $m = 2, a_6 = \frac{3}{5} a_4 = -\frac{1}{5} a_0$, etc.

Substitution in (i) gives

$$y = a_1 x + a_0 \left(1 - x^2 - \frac{1}{3} x^4 - \frac{1}{5} x^6 - \frac{1}{7} x^8 + \dots \right)$$

Which is the required series solution of the given differential equation.

Example 3: Find the first four terms in each portion of the series solution around

$x_0 = 0$ for the following differential equation $(1 + x^2) \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + 6y = 0 \dots (1)$

Solution: Here $p(x) = (1 + x^2)$, $p(0) = 1 \neq 0$.

So $x_0 = 0$ is an ordinary point for this differential equation.

$$y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$y'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{and} \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} \dots \dots \dots (2)$$

Putting above equation in (1)

$$\begin{aligned} (1 + x^2) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 4x \sum_{n=1}^{\infty} n a_n x^{n-1} + 6 \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=2}^{\infty} n(n-1) a_n x^n + \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=1}^{\infty} 4n a_n x^n + \sum_{n=0}^{\infty} 6a_n x^n &= 0 \\ \sum_{n=2}^{\infty} n(n-1) a_n x^n + \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=1}^{\infty} 4n a_n x^n + \sum_{n=0}^{\infty} 6a_n x^n &= 0 \end{aligned}$$

At this point we could strip out some terms to get all the series starting at $n=2$, but that's actually more work than is needed. Let's instead note that we could start the third series at $n=0$ if we wanted to because that term is just zero. Likewise, the terms in the first series are zero for both $n=1$ and $n=0$ and so we could start that series at $n=0$. If we do this all the series will now start at $n=0$ and we can add them up without stripping terms out of any series.

$$\sum_{n=2}^{\infty} \{n(n-1)a_n + (n+2)(n+1)a_{n+2} - 4n a_n + 6a_n\}x^n = 0$$

$$\sum_{n=2}^{\infty} \{(n^2 - 5n + 6)a_n + (n+2)(n+1)a_{n+2}\}x^n = 0$$

$$\sum_{n=2}^{\infty} \{(n-2)(n-3)a_n + (n+2)(n+1)a_{n+2}\}x^n = 0$$

$$(n-2)(n-3)a_n + (n+2)(n+1)a_{n+2} = 0, \quad n = 0, 1, 2, \dots$$

$$a_{n+2} = -\frac{(n-2)(n-3)a_n}{(n+2)(n+1)} \quad n = 0, 1, 2, \dots$$

$$n = 0, a_2 = -3a_0$$

$$n = 1, a_3 = -\frac{1}{3}a_0$$

$$n = 2, a_4 = 0$$

$$n = 3, a_5 = 0$$

$$y(x) = a_0(1 - 3x^2) + a_1(x - \frac{1}{3}x^3)$$

H.W. Examples.

1. Solve the equation $y'' = y'$ by the power series method.

2. Find a power series solution in powers of x of $y' + 2xy = 0$.

Frobenius Method for Solution near a Regular Singular Point:

Just as the power series method, the Frobenius method is useful for solving second order differential equations with variable coefficients about a regular singular point of the equation.

$$P_0(x) \frac{d^2y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x)y = 0 \dots \dots \dots (1) \quad \text{OR} \quad \frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$$

$$\text{where } P(x) = \frac{P_1(x)}{P_0(x)} \text{ and } Q(x) = \frac{P_2(x)}{P_0(x)}$$

Method of Solution

Let $x = 0$ be a regular singular point of equation (1), its solution can be represented in the form

$$y = x^m \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+m} = x^m (a_0 + a_1 x + a_2 x^2 + \dots), \quad a_0 \neq 0 \dots \dots \dots (2)$$

$$\text{Then, } \frac{dy}{dx} = \sum_{k=0}^{\infty} (k+m) a_k x^{k+m-1} \quad \text{and} \quad \frac{d^2y}{dx^2} = \sum_{k=0}^{\infty} (k+m)(k+m-1) a_k x^{k+m-2},$$

Substitute the value of y , $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in (1)

$$\begin{aligned} \therefore [m(m-1) a_0 x^{m-2} + (m+1)m a_1 x^{m-1} + (m+2)(m+1) a_2 x^m + \dots] \\ + [p_0 x + p_1 x^2 + \dots] \cdot [m a_0 x^{m-1} + (m+1) a_1 x^m + (m+2) a_2 x^{m+1} + \dots] \\ + [q_0 + q_1 x + \dots] \cdot [a_0 x^m + a_1 x^{m+1} + a_2 x^{m+2} + \dots] = 0 \end{aligned}$$

Equate to zero the coefficient of lowest power of x . This gives quadratic equation in m , which is called the indicial equation of the differential equation (1). Equate to zero the coefficients of various powers of x and express $a_1, a_2, a_3 \dots$ in terms of a_0 .

Substitute the values of a_1, a_2, a_3 in (2) to get solution of (1) having a_0 as arbitrary constant. One of the two solutions will always be the form (2), where m is a root of an indicial equation.

Let m_1 and m_2 be the roots of an indicial equation, then we have the following

- **Distinct roots not differing by an integer.**

$m_1 \neq m_2$ and $m_1 - m_2$ is not an integer. The general solution is

$$y = c_1(y)_{m_1} + c_2(y)_{m_2}$$

- **Double roots (Repeated roots):**

$m_1 = m_2 = b$. Then the general solution is

$$y = c_1(y)_1 + c_2(y)_2$$

$$y = c_1(y)_{m=b} + c_2 \left(\frac{\partial y}{\partial m} \right)_{m=b}$$

- **Roots differing by an integer** $m_1 - m_2 =$ an integer $m_1 < m_2$. In this case solutions corresponding to m_1 and m_2 may or may not be linearly independent.

EXAMPLE:-1: Solve in series the differential equation

$$4x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + y = 0$$

Solution: Clearly $x = 0$ is a regular singular point.

$$y = x^m \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} a_k x^{k+m} = x^m (a_0 + a_1 x + a_2 x^2 + \dots) \dots \dots (1)$$

be the series solution of given equation.

$$\text{Then } \frac{dy}{dx} = \sum_{k=0}^{\infty} (k+m) a_k x^{k+m-1}, \quad \frac{d^2y}{dx^2} = \sum_{k=0}^{\infty} (k+m)(k+m-1) a_k x^{k+m-2}$$

substituting the values of $y, \frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in the given equation, we get

$$4x \sum_{k=0}^{\infty} (k+m)(k+m-1) a_k x^{k+m-2} + 2 \sum_{k=0}^{\infty} (k+m) a_k x^{k+m-1} + \sum_{k=0}^{\infty} a_k x^{m+k} = 0$$

$$\begin{aligned} \text{or } 4x[m(m-1)a_0x^{m-2} + (m-1)ma_1x^{m-1} + (m+2)(m+1)a_2x^m + (m+3)(m+2)a_3x^{m+1} + \dots] \\ + 2[ma_0x^{m-1} + (m+1)a_1x^m + (m+2)a_2x^{m+1} + (m+3)a_3x^{m+2} + \dots] \\ + [a_0x^m + a_1x^{m+1} + a_2x^{m+2} + a_3x^{m+3} + \dots] = 0 \end{aligned}$$

The lowest power of x is x^{m-1} . Equating to zero the coefficient of x^{m-1} , we get

$$a_0[4m(m-1) + 2m] = 0,$$

Its roots are $m = 0$ and $m = \frac{1}{2}$, which are distinct and not differing by an integer.

Equating the coefficient of x^{k+m} , gives

$$a_{k+1} = -\left(\frac{a_k}{(2m+2k+2)(2m+2k+1)}\right), k = 0, 1, 2, 3, \dots$$

First solution:-The solution corresponding to $m = \frac{1}{2}$ is obtained from the recurrence relation.

$$a_{k+1} = -\left(\frac{a_k}{(2k+3)(2k+2)}\right)$$

Hence,

$$a_1 = -\frac{a_0}{3.2}, a_2 = -\frac{a_1}{5.4}, a_3 = -\frac{a_2}{7.6}, \text{etc},$$

$$\therefore a_1 = -\frac{a_0}{3!}, a_2 = \frac{a_0}{5!}, a_3 = -\frac{a_0}{7!} \dots \dots$$

and in general $a_n = \frac{(-1)^n}{(2n+1)!}$ as $(n = 0, 1, 2, 3 \dots)$, The first solution is

$$y_1(x) = a_0 x^{\frac{1}{2}} \left(1 - \frac{1}{6}x + \frac{1}{120}x^2 - \dots \dots \right)$$

Second solution:-The solution corresponding to $m = 0$ is obtained from the recurrence relation.

$$a_{k+1} = -\left(\frac{a_k}{(2k+3)(2k+2)} \right)$$

Hence,

$$a_1 = -\frac{a_0}{2.1}, a_2 = -\frac{a_1}{4.3}, a_3 = -\frac{a_2}{6.5}, \text{etc},$$

$$\therefore a_1 = -\frac{a_0}{2!}, a_2 = \frac{a_0}{4!}, a_3 = -\frac{a_0}{6!} \dots \dots$$

and in general $a_n = \frac{(-1)^n}{2n!}$ as $(n = 0, 1, 2, 3 \dots)$,

The second solution is

$$y_2 = a_0 \left(1 - \frac{x}{2} + \frac{x^2}{24} - \dots \dots + \dots \right),$$

Hence general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= Ax^{\frac{1}{2}} \left(1 - \frac{1}{6}x + \frac{1}{120}x^2 - \dots \dots \right) + B \left(1 - \frac{x}{2} + \frac{x^2}{24} - \dots \dots + \dots \right),$$

where $A = c_1 a_0$ and $B = c_2 a_0$

H.W.

EXAMPLE-3: Solve in series the differential equation $x^2 y'' + xy' + (x^2 - 4)y = 0$

LEGENDRE POLYNOMIALS

$$P_n(x) = \sum_{r=0}^N (-1)^r \frac{(2n-2r)!}{2^n r! (n-r)! (n-2r)!} x^{n-2r}$$

Where, $N = \frac{n}{2}$, when n is even, $N = \frac{n-1}{2}$ when n is odd

From 2 we get the following set of polynomials:

$$\begin{aligned} P_0(x) &= 1, & P_1(x) &= x, \\ P_2(x) &= \frac{1}{2}(3x^2 - 1), & P_3(x) &= \frac{1}{2}(5x^3 - 3x), \\ P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3), & P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x), \text{ etc.} \end{aligned}$$

Each satisfies a Legendre differential equation in which n has the value indicated by the subscript.

Note: Rodrigue's Formula

$$P_n(x) = \frac{1}{2^n (n!)} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Example 1: show that $x^3 = \frac{2}{5}P_3(x) + \frac{3}{5}P_1(x)$.

Solution: we know that

$$P_n(x) = \frac{1}{2^n (n!)} \frac{d^n}{dx^n} (x^2 - 1)^n \dots \dots \dots (1)$$

Putting, $n = 1$ and $n = 3$ in 1 we get,

$$P_1(x) = \frac{1}{2} \frac{d}{dx} (x^2 - 1) = x$$

$$\begin{aligned} P_3(x) &= \frac{1}{2^3 (3!)} \frac{d^3}{dx^3} (x^2 - 1)^3 \\ &= \frac{1}{48} \frac{d^3}{dx^3} (x^6 - 3x^4 + 3x^2 - 1) = \left[\frac{1}{2} (5x^3 - 3x) \right] \end{aligned}$$

$$\therefore \frac{2}{5}P_3(x) + \frac{3}{5}P_1(x) = \frac{2}{5} \left[\frac{1}{2} (5x^3 - 3x) \right] + \frac{3}{5}(x) = \frac{1}{5}(5x^3 - 3x) + \frac{3}{5}x = x^3$$

Example 2: show that $x^4 = \frac{1}{35} [8P_4(x) + 20P_2(x) + 7P_0(x)]$.

Solution: we know that

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^3 + 3)$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1), \quad P_0(x) = 1.$$

$$\therefore \frac{1}{35}[8P_4(x) + 20P_2(x) + 7P_0(x)] = \frac{1}{35}[(35x^4 - 30x^3 + 3) + 10(3x^2 - 1) + 7] = x^4$$

Example 3: Express $f(x) = x^4 + 3x^3 - x^2 + 5x - 2$ in terms of Legendre's polynomials.

BESSEL FUNCTION:

The differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0 \text{ --- (1)}$$

Is called **Bessel's equation** of order n and its particular solutions are called **Bessel functions** of order n .

Application: In vibration problems, electric fields, heat conduction, fluid flow.

A) Bessel Function of the first kind of order n .

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k}$$

where n is a positive real number or zero.

B) Bessel function of the first kind of order $-n$.

$$J_{-n}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{-n+2k}$$

Hence the complete solution of **Bessel's equation** (1) may be expressed as

$$y = AJ_n(x) + BJ_{-n}(x)$$

When A and B are arbitrary constants.

C) Bessel equation the first kind of order zero.

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k}$$

$$\text{since } \Gamma(k+1) = k!$$

EXAMPLE 1 : Bessel functions $J_0(x)$ and $J_1(x)$.

Solution:

$$\text{Since, } J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k}$$

$$\text{So, } J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+1)} \left(\frac{x}{2}\right)^{2k}$$

$$= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \text{ as, } \Gamma(k+1) = k!$$

$$= 1 - \left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2} \left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2} \left(\frac{x}{2}\right)^6 + \dots$$

$$\therefore J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$J_1(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+2)} \left(\frac{x}{2}\right)^{1+2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (k+1)!} \left(\frac{x}{2}\right)^{2k}$$

$$= \frac{x}{2} - \frac{1}{2!} \left(\frac{x}{2}\right)^3 + \frac{1}{2! 3!} \left(\frac{x}{2}\right)^5 - \dots + \dots$$

$$\therefore J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} - \dots + \dots$$

In particular $J_0(0) = 1$ and $J_1(0) = 0$

EXAMPLE 2 : Prove that $\therefore J'_0(x) = -J_1(x)$

Solution: we have

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$\text{and } J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} - \dots + \dots$$

$$\text{Now } \frac{d}{dx} J_0(x) = J'_0(x) = 0 - \frac{x}{2} + \frac{x^3}{2^2 \cdot 4} - \frac{x^5}{2^2 \cdot 4^2 \cdot 6} + \dots$$

$$= -\left(\frac{x}{2} - \frac{4x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} + \dots\right)$$

$$\therefore J'_0(x) = -J_1(x)$$

Prove that

$$(i) J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad (ii) J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

Solution: (i)

$$\begin{aligned} J_{\frac{1}{2}}(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma\left(k + \frac{3}{2}\right)} \left(\frac{x}{2}\right)^{\left(\frac{1}{2}+2k\right)} \\ &= \frac{x^{\left(-\frac{1}{2}\right)}}{2^{\left(-\frac{1}{2}\right)}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2^{2k+1} k! \Gamma\left(k + \frac{3}{2}\right)} \\ &= \frac{x^{\left(-\frac{1}{2}\right)}}{2^{\left(-\frac{1}{2}\right)}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2^{2k+1} k! \Gamma\left(k + \frac{3}{2}\right)} \\ &= \sqrt{\frac{2}{x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2^k k! \left(k + \frac{1}{2}\right) \dots \Gamma\left(\frac{1}{2}\right)} \\ &= \sqrt{\frac{2}{x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2^k k! (2k+1)(2k-1) \dots \sqrt{\pi}} \\ &= \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = \sqrt{\frac{2}{\pi x}} \left(\frac{x}{1!} - \frac{x^3}{3!} + \dots \right) \Rightarrow J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x \end{aligned}$$

$$\begin{aligned} (ii) J_{-\frac{1}{2}}(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma\left(k - \frac{1}{2} + 1\right)} \left(\frac{x}{2}\right)^{\left(-\frac{1}{2}+2k\right)} \\ &= \frac{x^{\left(-\frac{1}{2}\right)}}{2^{\left(-\frac{1}{2}\right)}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^{2k} k! \Gamma\left(k + \frac{1}{2}\right)} \\ &= \sqrt{\frac{2}{x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^{2k} k! \left(k - \frac{1}{2}\right) \dots \Gamma\left(\frac{1}{2}\right)} \\ &= \sqrt{\frac{2}{x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^{2k} k! (2k-1) \dots \sqrt{\pi}} \end{aligned}$$

$$= \sqrt{\frac{2}{\pi x}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

$$= \sqrt{\frac{2}{\pi x}} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) \Rightarrow J_{\frac{1}{-2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

PROPERTIES OF BESSEL FUNCTION

Recurrence relations for J_n :

1. $J_n'(x) + \left(\frac{n}{x}\right)J_n(x) = J_{n-1}(x)$
2. $J_n'(x) - \left(\frac{n}{x}\right)J_n(x) = -J_{n+1}(x)$
3. $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$
4. $\left(\frac{2n}{x}\right)J_n(x) = J_{n-1}(x) + J_{n+1}(x)$